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Fluxtools: An R Package with an Interactive Shiny App for Reproducible QA/QC of Eddy Covariance Data Aligned with AmeriFlux Standards

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Software

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Summary

Eddy covariance data processing requires extensive quality control (QA/QC) to identify and remove implausible or erroneous half-hourly flux data before submission to public data repositories such as AmeriFlux (AmeriFlux Management Project, 2025). *Fluxtools* (Key, 2025) is an R ($\geq$ 4.5.0; (2025)) *Shiny* (Chang et al., 2024) application built with *Plotly* (Sievert et al., 2024) and *dplyr* (Wickham et al., 2023) packages designed to streamline this workflow by providing interactive visualization, year-based filtering, and on-the-fly R code generation for specified data removal. Users can visually flag anomalous data points (i.e., periods of sensor failure, physically implausible data), accumulate multiple cleaning steps, inspect R^2 values before and after data cleaning via base R's *lm()* function, and export a zipped folder containing a cleaned .csv file and a full R script that records every decision. *Fluxtools* significantly accelerates the QA/QC workflow, ensuring transparent, reproducible, and shareable data cleaning suitable for final dataset preparation and repository submission.

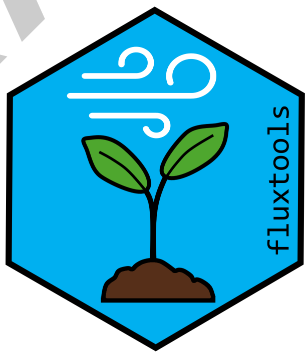

Figure 1: *Fluxtools* hex logo

Key features:

- **Interactive Plotly Scatterplots**: Plot any numeric or time variable; hover mouse over data points to see timestamps and values; export plots as .png directly from the app
- **Flexible point selection**: Box-select, lasso, or apply standard-deviation (σ) cutoffs to mark selected points. *Fluxtools* automatically generates removal ready-to-copy R snippets (*dplyr::case_when(… ~ NA)*) in the *Current* code pane. *See Fig 2* for interface and data selection example
- **On-the-fly R code generation**: After point selection, clicking *Flag Data* automatically

highlights chosen points in yellow and appends the corresponding removal code into the *Accumulated* code panel for easy and continuous data selection

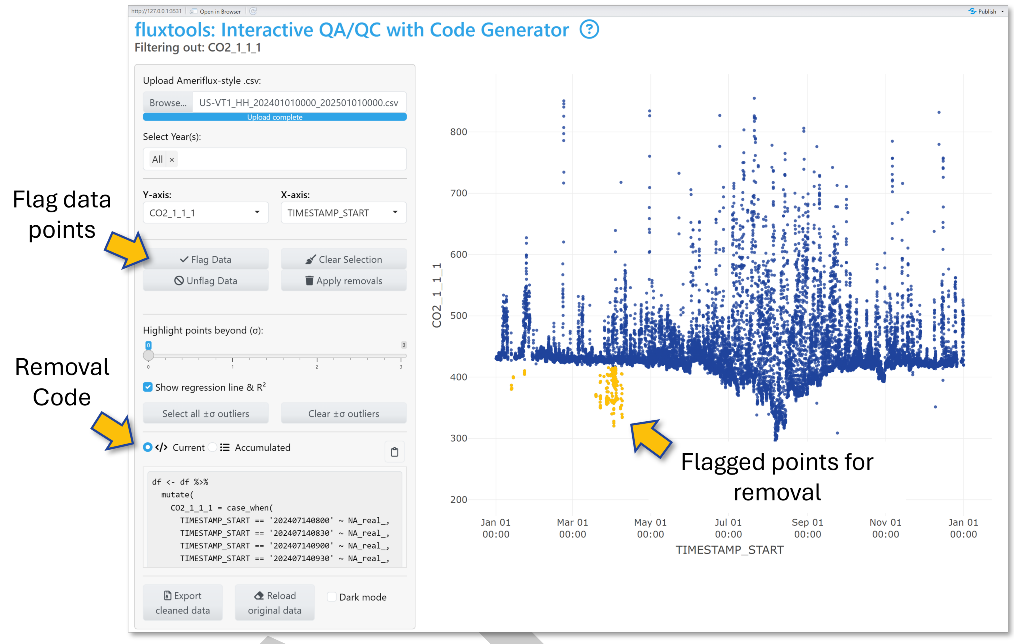

Figure 2: Example of the *Fluxtools* interface and data selection

- **Before/after R² diagnostics**: For any numeric variable comparison, *Fluxtools* fits a linear regression model and reports its R² value. Selecting points re-computes R² as if those points were removed, allowing for easy comparison. *Fig 3* shows this process in *Fluxtools* using the *$\pm\sigma$ outliers selection tool:* The top (red) R² uses all data, while the bottom R² (orange) omits selected points from the linear regression

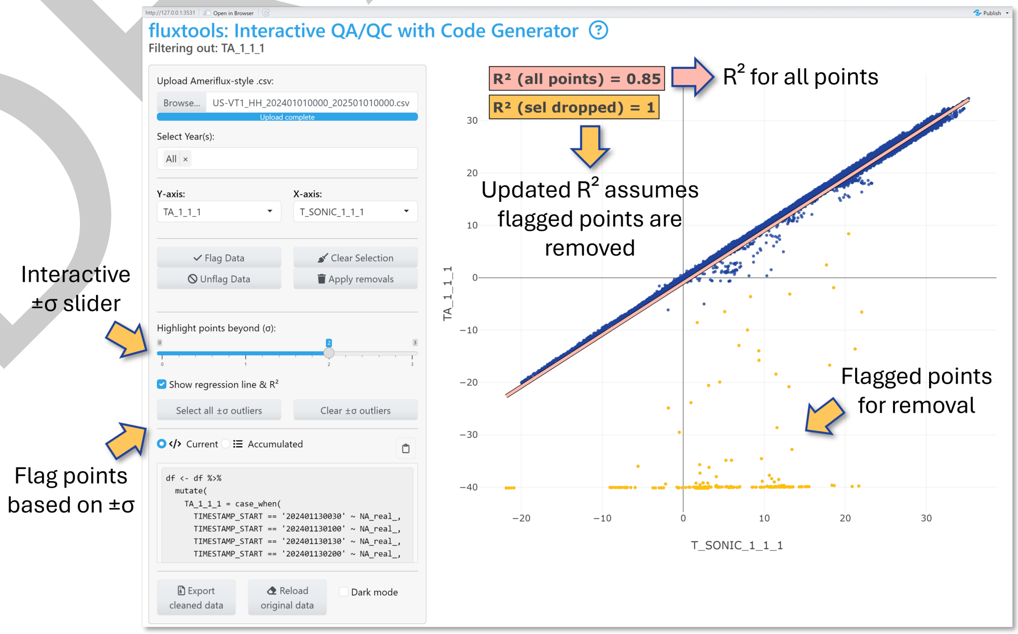

Figure 3: Example of R² diagnostics using the $\pm\sigma$ outliers cutoffs and selection

- **Export cleaned .csv file and R script**: *Apply removals* in-app (converting data points into *NA*s for selected timestamps) then *Export cleaned data* to download a cleaned .csv file and a comprehensive R script documenting each data removal step

Statement of need

High-frequency (10Hz; data recorded 10x per second) eddy covariance measurements generate large datasets that must be aggregated into half-hourly fluxes, using careful QA/QC (Burba, 2021). At this high temporal resolution, intermittent periods of sensor drift or failure are common, making manual data cleaning an integral part of the workflow.

Tools like *Loggernet's CardConvert* feature (Campbell Scientific, Inc., 2025) convert raw 10Hz eddy covariance data into half-hourly intervals, preparing them for flux estimation. Software like *EddyPro* (LI-COR Biosciences, 2021) then computes turbulent fluxes of CO_2, H_2O, and energy using these half-hourly inputs. Post-processing *R* packages like *REddyProc* (Wutzler et al., 2024), and Python tools like PyFluxPro (Isaac, 2021), automate u*-threshold filtering, gap-filling, and flux partitioning. These tools excel at bulk data processing but offer no interactive means to inspect or carefully remove outliers that require a human eye.

In practice, data managers resort to custom scripts, manual visualization, and fragmented documentation to detect and remove erroneous data points caused by sensor drift, malfunction, or calibration issues. These procedures are labor-intensive, error-prone, difficult to reproduce, and lack transparency. *Fluxtools* addresses these challenges by pairing an interactive scatterplot-based interface with on-the-fly R code generation. Users can visually flag implausible half-hourly data points, automatically generate or directly apply the exact *case_when(… ~ NA) dplyr* code snippets and export a .zip file containing a cleaned .csv file plus a comprehensive R script documenting each QA/QC decision.

By combining interactivity with code-based reproducibility, *Fluxtools* streamlines QA/QC workflows by enhancing transparency, reducing manual effort, and accelerating the preparation of flux data for repository submissions (e.g., Ameriflux). Ultimately, *Fluxtools* lowers the barriers to robust and reproducible QA/QC, allowing data managers and scientists to spend less time on manual anomaly detection and more on scientific analysis and interpretation.

Code Example

Installation

```r
# Option 1.) Install from CRAN
install.packages("fluxtools")
library(fluxtools)

# Option 2.) Install most recent version from GitHub
library(remotes)
remotes::install_git("https://github.com/kesondrakey/fluxtools.git")
```

Call the Fluxtools App

```r
library(fluxtools)

#Set your site's UTC offset (e.g., -5 for Eastern Standard Time)
run_flux_qaqc(-5)
```

Acknowledgments

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